SI 315

Models of Social Information Theory

Silas Falde

October 13, 2026

Logistics

- No AI-generated submissions to assignments
- Required attendance at both lectures and discussions
- Two exams, final is non-cumulative
- iClicker both participation and correctness
- Lowest two grades dropped from each of assignments, quizzes, discussion attendance, lecture attendance

Contents

1 Game Theory **2**

1.1 Strategic-Form Games . . . 2
1.2 Dominant Strategies . . . 2
1.3 Example Games . . . 2
1.3.1 Presentation vs. Exam . . . 2
1.3.2 Prisoner's Dilemma . . . 2
1.3.3 A New Product . . . 2
1.3.4 Three Clients . . . 3
1.4 Nash Equilibrium . . . 3
1.5 Focal Points . . . 3
1.6 Mixed Strategies . . . 3
1.6.1 Finding a Mixed-Strategy Equilibrium . . . 3
1.6.2 Solving for an Unknown Payoff . . . 4
1.6.3 Probabilities from a Complete Payoff Matrix . . . 4
1.7 Matching Pennies . . . 5
1.8 Coordination Game . . . 5
1.9 Network Traffic . . . 5
1.10 Auctions . . . 6
1.11 Bipartite Matching . . . 6
1.11.1 Weighted Edges . . . 6
1.11.2 Finding Market-Clearing Prices . . . 6

1 Game Theory

1.1 Strategic-Form Games

- Game: players, available strategies, and payoffs for each strategy profile
- Strategy profile: one strategy chosen by each player
- Payoff: value or outcome a player receives from a strategy profile
- Best response: strategy that maximizes a player's payoff given the other players' strategies
- Strict best response: strategy that gives a strictly higher payoff than every alternative, given the other players' strategies
- More than one best response possible when strategies tie for the highest payoff

1.2 Dominant Strategies

- Dominant strategy: best response to every possible strategy profile of the other players
- Strictly dominant strategy: gives a strictly higher payoff than every alternative for every possible strategy profile of the other players
- Strictly dominant strategy: rational choice regardless of what the other players do
- A game may have no dominant strategy, or may have dominant strategies for some or all players

1.3 Example Games

1.3.1 Presentation vs. Exam

- Compare each player's available choices and payoffs in every outcome
- Mark each player's best response to each choice of the other player
- Use the payoff matrix to determine whether choices align or conflict

1.3.2 Prisoner's Dilemma

- Two players independently choose to cooperate or defect
- Defecting: strictly dominant strategy for each player
- (Defect, Defect): Nash equilibrium, despite mutual cooperation giving both players a better outcome
- Individual incentives can produce an outcome worse for both players than cooperation

1.3.3 A New Product

- Analyze the available choices and payoffs for each player using the lecture payoff matrix
- Check how one player's best response changes with the other player's choice
- Use best responses to identify any Nash equilibria

1.3.4 Three Clients

- Identify the players, available strategies, and payoffs from the lecture example
- For each player, compare payoffs across their choices while holding the other players' choices fixed
- Mark mutual best responses to locate Nash equilibria

1.4 Nash Equilibrium

- Two-player strategy profile (S, T): Nash equilibrium if S is a best response to T and T is a best response to S
- General case: every player's strategy is a best response to the strategies chosen by all other players
- At equilibrium: no player can improve their payoff by changing strategy alone
- Finite-game existence theorem: every game with finitely many players and finitely many strategies has at least one Nash equilibrium in mixed strategies
- Pure-strategy Nash equilibrium: equilibrium using a single strategy per player; not guaranteed to exist

1.5 Focal Points

- Focal point: salient outcome that helps players coordinate without communication
- Salience may come from conventions, shared expectations, or a distinctive feature of the game
- Focal points can help players select among multiple coordination-game equilibria

1.6 Mixed Strategies

- Mixed strategy: probability distribution over a player's available pure strategies
- Pure strategy: one action chosen with probability 1
- Expected payoff: probability-weighted average payoff across possible outcomes
- For strategy profile (s_i, s_{-i}) drawn from mixed strategies:

$$\mathbb{E}[u_i] = \sum_{s_i, s_{-i}} \Pr(s_i) \Pr(s_{-i}) u_i(s_i, s_{-i})$$

- Mixed-strategy Nash equilibrium: each player's probability distribution is a best response to the others' distributions
- In equilibrium, every pure strategy used with positive probability gives the player the same expected payoff

1.6.1 Finding a Mixed-Strategy Equilibrium

- Assign probabilities to one player's strategies
- Choose probabilities that make the other player indifferent among their pure strategies
- Repeat in the opposite direction to find the first player's equilibrium probabilities
- Indifference makes mixing optimal; a player would otherwise put all probability on a higher-payoff pure strategy

1.6.2 Solving for an Unknown Payoff

- **Indifference principle**: in a mixed-strategy NE, each player's mix makes the *other* player indifferent among strategies played with positive probability
- Steps:
  1. Get equilibrium probabilities (usually given)
  2. Write indifference condition for the player whose payoffs are known: uses the opponent's mix
  3. Write indifference condition for the player with the unknown payoff x: $\mathbb{E}[u_1(U)] = \mathbb{E}[u_1(D)]$
  4. Solve for x: linear, one equation, one unknown
- Unknown payoff of player i appears in i's indifference condition, which uses the *opponent's* probabilities
- If x also affects the opponent's mix: use both indifference equations with given probabilities
- If no probabilities given: treat p, q as unknowns; two indifference equations, solve for p, q, x
- Check: all probabilities in $[0, 1]$; no profitable deviation to a pure strategy outside the mix

Example

- Payoffs (Player 1, Player 2):

	L	R
U	$x, 2$	$0, 1$
D	$1, 0$	$3, 3$

- Given: Player 2 plays L with probability $q = \frac{3}{4}$
- Player 1 indifferent between U and D:

$$\mathbb{E}[u_1(U)] = qx + (1-q)\cdot 0 = \tfrac{3}{4}x$$
$$\mathbb{E}[u_1(D)] = q\cdot 1 + (1-q)\cdot 3 = \tfrac{3}{4} + \tfrac{3}{4} = \tfrac{3}{2}$$

- Set equal: $\frac{3}{4}x = \frac{3}{2}$, so $\boxed{x = 2}$

1.6.3 Probabilities from a Complete Payoff Matrix

- p = Pr(Player 1 plays U); q = Pr(Player 2 plays L)
- p comes from **Player 2's** indifference; q comes from **Player 1's** indifference
- Own payoffs only determine what the opponent does; given q is not needed to find p
- Check dominated strategies first: strictly dominated gets probability 0; fully mixed NE needs no dominant strategy
- Valid only if $0 < p, q < 1$

Example

- Payoffs (Player 1, Player 2):

	L	R
U	$4, 3$	$1, 1$
D	$2, 0$	$5, 4$

- Find p: Player 2 indifferent between L and R

$$\mathbb{E}[u_2(L)] = 3p + 0(1-p) = 3p$$
$$\mathbb{E}[u_2(R)] = 1p + 4(1-p) = 4 - 3p$$

- $3p = 4 - 3p$, so $\boxed{p = \frac{2}{3}}$
- Find q: Player 1 indifferent between U and D

$$\mathbb{E}[u_1(U)] = 4q + 1(1-q) = 1 + 3q$$
$$\mathbb{E}[u_1(D)] = 2q + 5(1-q) = 5 - 3q$$

- $1 + 3q = 5 - 3q$, so $\boxed{q = \frac{2}{3}}$
- Mixed NE: $p = q = \frac{2}{3}$
- Sanity check: $q = \frac{1}{2}$ is not an equilibrium value; $\mathbb{E}[u_1(U)] = 2.5 < 3.5 = \mathbb{E}[u_1(D)]$, so Player 1 strictly prefers D

1.7 Matching Pennies

- Each player chooses Heads or Tails
- Player 1 wins when choices match; Player 2 wins when choices differ
- No pure-strategy Nash equilibrium: either player can switch and improve their payoff
- Let q be the probability Player 2 chooses Heads
- Player 1's expected payoff from Heads: $q - (1-q) = 2q - 1$
- Player 1's expected payoff from Tails: $(1-q) - q = 1 - 2q$
- Player 1 is indifferent when $2q - 1 = 1 - 2q$, so $q = \frac{1}{2}$
- By symmetry, Player 1 also chooses Heads with probability $\frac{1}{2}$
- Mixed-strategy Nash equilibrium: each player chooses Heads and Tails with probability $\frac{1}{2}$

1.8 Coordination Game

- Players benefit from choosing strategies that match or coordinate
- Coordination games often have multiple Nash equilibria
- Equilibrium selection: players need expectations, conventions, or focal points to coordinate on one equilibrium
- A coordinated outcome may be better for everyone than a mismatched outcome

1.9 Network Traffic

- Players choose routes in a network to minimize their own travel time
- Travel time on a route depends on the number of players using it
- Nash equilibrium occurs when no player can reduce their travel time by unilaterally changing routes
- Equilibria may be inefficient compared to a centrally coordinated solution (Braess's paradox)

1.10 Auctions

- Types of auctions: ascending (English), descending (Dutch), first-price sealed-bid, second-price sealed-bid (Vickrey)
- Second-price sealed-bid auction: buyers are players, bids are strategies—assume each buyer has a true value v_i and submits a bid b_i; highest bidder "payoff" is $v_i - b_j$ (the money saved compared to their true value), where b_j is the second-highest bid; dominant strategy is to bid their true value $b_i = v_i$; truthful bidding leads to efficient allocation of the item.

1.11 Bipartite Matching

- Bipartite graph: graph whose vertices can be divided into two disjoint sets where no two vertices within the same set are adjacent
- Bipartite matching: set of edges in a bipartite graph such that no two edges share a common vertex
- Maximum bipartite matching: bipartite matching that contains the largest possible number of edges
- Constricted set: subset of vertices in one partition of a bipartite graph such that the number of neighbors in the other partition is strictly smaller than the number of vertices in the subset

1.11.1 Weighted Edges

- Weighted bipartite graph: bipartite graph where each edge has an associated weight
- Maximum weight bipartite matching (optimal assignment): bipartite matching that maximizes the sum of the weights of the selected edges

1.11.2 Finding Market-Clearing Prices

- Market-clearing prices: prices assigned to edges in a weighted bipartite graph such that each vertex in one partition is matched to a vertex in the other partition with the highest possible weight given the prices, ensuring that the market clears (all supply meets demand); aka, competitive equilibrium prices.
- Iterative procedure to find market-clearing prices:

  1. Initialize prices for all edges, typically starting with zero.
  2. Construct preferred-seller graph based on the current prices, where each vertex in one partition points to its most preferred vertex in the other partition.
  3. Find a matching in the preferred-seller graph.
  4. If the matching is not perfect, increase the prices of the nodes with multiple preferred edges and repeat the process until a perfect matching is found.